Review: Doubly Robust Estimation of an ATE

AIPW, Influence Functions, and Neyman Orthogonality

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Three Baselines, Three Defects

A GOTV field experiment assigns canvass $D_i \in \{0,1\}$ with $\Pr(D_i = 1) = p$, independent of potential outcomes $(Y_i(0), Y_i(1))$ and covariates X_i. Target:

$$\tau = \mathbb{E}[Y(1) - Y(0)].$$

Three consistent estimators, each with a single weakness:

- **Difference in means.** $\hat{\tau}_{DM} = \bar{Y}_1 - \bar{Y}_0$. Uses randomization, ignores everything about X_i.
- **Outcome regression.** $\hat{\tau}_{OR} = \frac{1}{n}\sum_i \left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\right]$. Consistent only if $\hat{\mu}_d \to \mu_d(x) = \mathbb{E}[Y(d) \mid X = x]$.
- **Inverse-probability weighting.** $\hat{\tau}_{IPW} = \frac{1}{n}\sum_i \left[\frac{D_i Y_i}{p} - \frac{(1-D_i)Y_i}{1-p}\right]$. Reweights observed arms, but throws away the covariate structure.

AIPW splices the regression and the weighting pieces together and inherits the strengths of both.

The AIPW Estimator

Definition (Augmented IPW)

With $\hat{\mu}_d(\cdot)$ a fitted outcome model (sample-split, so independent of the units in the sum),

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n}\sum_{i=1}^{n}\left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \tfrac{D_i}{p}\big(Y_i - \hat{\mu}_1(X_i)\big) - \tfrac{1-D_i}{1-p}\big(Y_i - \hat{\mu}_0(X_i)\big)\right].$$

Two readings of the same formula.

- *Outcome regression with an IPW correction.* Average the predicted treatment effect; the remaining terms are mean-zero residuals that correct for misspecification of $\hat{\mu}_d$.
- *IPW with an outcome-regression control variate.* Subtract $\hat{\mu}_d(X_i)$ as a baseline whose expectation does not shift the target, but whose variation absorbs the predictable part of Y_i.

Influence Function Representation

Every $\sqrt{n}$-consistent, asymptotically normal estimator admits a representation as a sample average of an i.i.d. mean-zero random variable plus a vanishing remainder:

$$\sqrt{n}(\hat{\tau} - \tau) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi(Z_i) + o_p(1).$$

$\psi(Z)$ is the **influence function**. Its variance is the asymptotic variance of $\hat{\tau}$.

For AIPW evaluated at the true μ_d,

$$\psi(Z) = \mu_1(X) - \mu_0(X) - \tau + \tfrac{D}{p}\big(Y - \mu_1(X)\big) - \tfrac{1-D}{1-p}\big(Y - \mu_0(X)\big).$$

ψ is the GMM moment function for the scalar parameter τ: $\hat{\tau}$ solves the just-identified moment condition $n^{-1}\sum_i \psi(Z_i; \tau) = 0$, which is what connects this construction back to GMM.

CLT for the Infeasible Target

Let $\hat{\tau}_\star$ be the AIPW formula evaluated at the truth μ_d rather than at $\hat{\mu}_d$:

$$\hat{\tau}_\star = \frac{1}{n}\sum_i \left[\mu_1(X_i) - \mu_0(X_i) + \tfrac{D_i}{p}(Y_i - \mu_1(X_i)) - \tfrac{1-D_i}{1-p}(Y_i - \mu_0(X_i))\right].$$

The summands are i.i.d. with mean τ, so the standard CLT gives

$$\boxed{\sqrt{n}(\hat{\tau}_\star - \tau) \xrightarrow{d} N(0, \mathrm{Var}(\psi(Z))).}$$

Why this is not enough. $\hat{\tau}_\star$ is infeasible; the true μ_d is unknown. We need to transfer the limit to the feasible $\hat{\tau}_{\mathrm{AIPW}}$, which means controlling the gap caused by $\hat{\mu}_d - \mu_d$. That step is what Neyman orthogonality buys.

Neyman Orthogonality

Under randomization, $D \perp X$ with $\mathbb{E}[D] = p$, so for any deterministic $\Delta(\cdot)$,

$$\mathbb{E}\left[1 - \frac{D}{p} \,\middle|\, X\right] = 0 \implies \mathbb{E}\left[\left(1 - \frac{D}{p}\right)\Delta(X)\right] = 0.$$

Why this matters. A direct subtraction shows the gap between feasible and infeasible AIPW is

$$\hat{\tau}_{\text{AIPW}} - \hat{\tau}_{\star} = \frac{1}{n}\sum_i \left[\left(1 - \frac{D_i}{p}\right)(\hat{\mu}_1 - \mu_1)(X_i) - \left(1 - \frac{1-D_i}{1-p}\right)(\hat{\mu}_0 - \mu_0)(X_i)\right].$$

Each summand is conditionally mean zero by the orthogonality identity. Under sample splitting and L^2-consistency of $\hat{\mu}_d$, Chebyshev (HW4 Q7) bounds the remainder at $o_p(n^{-1/2})$, and Slutsky transfers the CLT to the feasible $\hat{\tau}_{\text{AIPW}}$.

Double Robustness

The orthogonality argument used the *true* propensity p. AIPW is stronger:

$$\hat{\tau}_{\mathrm{AIPW}} \textit{ is consistent if either } \hat{\mu}_d \textit{ or } \hat{p} \textit{ is correctly specified.}$$

Let $m_d(X)$ and $\tilde{p}$ be the probability limits of $\hat{\mu}_d$ and $\hat{p}$. Then

$$\operatorname{plim} \hat{\tau}_{\mathrm{AIPW}} = \mathbb{E}[m_1 - m_0] + \tfrac{p}{\tilde{p}}\big(\mathbb{E}[Y(1)] - \mathbb{E}[m_1]\big) - \tfrac{1-p}{1-\tilde{p}}\big(\mathbb{E}[Y(0)] - \mathbb{E}[m_0]\big).$$

Two ways this collapses to τ:

- $\tilde{p} = p$: both ratios equal 1, leaving $\mathbb{E}[Y(1) - Y(0)] = \tau$.
- $\mathbb{E}[m_d] = \mathbb{E}[Y(d)]$: each correction term vanishes, leaving the regression piece equal to τ.

For marginal estimands like the ATE, the second condition is weaker than "correct outcome model". Even $\hat{\mu}_d \equiv \bar{Y}_d$ satisfies it: wrong about $\mu_d(x)$, but right about $\mathbb{E}[Y(d)]$.

Variance, Efficiency, Inference

Asymptotic variance. By the law of total variance (condition on X, then on (X, D)),

$$\mathrm{Var}(\psi(Z)) = \mathbb{E}\left[\frac{\sigma_1^2(X)}{p} + \frac{\sigma_0^2(X)}{1-p}\right] + \mathrm{Var}\big(\mu_1(X) - \mu_0(X)\big),$$

with $\sigma_d^2(x) = \mathrm{Var}(Y(d) \mid X = x)$. This is the semiparametric efficiency bound for τ (Hahn 1998).

Comparison to difference-in-means. Under Bernoulli randomization, $n\,\mathrm{Var}(\hat{\tau}_{DM}) \to \mathrm{Var}(Y(1))/p + \mathrm{Var}(Y(0))/(1-p)$. Use $\mathrm{Var}(Y(d)) = \mathbb{E}[\sigma_d^2(X)] + \mathrm{Var}(\mu_d(X))$ to expand both sides; the gap collapses cleanly at $p = 1/2$.

Feasible inference. From slide 6, $\sqrt{n}(\hat{\tau}_{\mathrm{AIPW}} - \tau) \xrightarrow{d} N(0, \mathrm{Var}(\psi))$. Estimate $\mathrm{Var}(\psi)$ by the empirical second moment of the plug-in influence function $\hat{\psi}_i$; the 95% CI and the Wald statistic for $H_0 : \tau = \tau_0$ follow.